Sequencing data
Paired FASTQs

Auxiliary data
Metadata and sample manifest

Read QC

Cutadapt / fastp / VSEARCH / SeqKit

Optional paired-primer removal and amplicon-family audit
Read QC, paired-read merging, filtering and relabelling

Denoise and remove chimeras

VSEARCH: dereplication / UNOISE / UCHIME

Dereplicate and denoise reads; remove chimeras
Construct ASV sequences and the sample-by-ASV count matrix

Alignment and taxonomic labelling

SINA / QIIME 2 / SILVA

Align inferred sequences and assign full ASV taxonomy
Retain taxonomy confidence and reference provenance

Contamination and host removal

decontam / MITOMASTER / BLAST+ / SeqKit

Biological depth QC; TECH / BIO prevalence tests when enabled
Combine flagged ASVs; screen mitochondrial and contaminant references

Abundance / prevalence filtering and ambiguous taxon removal

Combined count-table and FASTA filtering

Apply configured abundance, prevalence and taxonomy rules
Remove zero-count features; synchronize final counts and FASTA; audit exclusions

Batch correction with metadata

ConQuR (optional) / metadata preparation

Join sample metadata; construct long / wide tables and CLR representations
Cohort diagnostics and selection; synchronize corrected and raw analytical inputs

Filtered and corrected ASVs / count data

Processed VOC data
(optional)

Input

Process

Output

ASV analysis

Diversity estimation

Shannon / Bray-Curtis / Jaccard / PERMANOVA

scikit-bio / vegan; ordination and collectors curves
Full biological cohort retained; matched subsets where configured

Clustering and outlier detection

UMAP / CLR / HDBSCAN

Embedding, group diagnostics and outlier summaries
Selected cohort feeds subsequent study-specific analyses

Indicator species analysis

indicspecies; permutation tests / FDR

Group and multigroup indicators; taxonomy-linked plots
Indicator membership supplies ASV-VOC overlays

Taxon clustering and mapping

Hierarchical clustering / heatmaps / vegan

ASV clustermaps, taxonomy / group tests and paired contrasts
Patient-count power simulations and study-design summaries

Correlation network analysis

SPIEC-EASI / igraph: Leiden / NetworkX

Filtered network inference, consensus modules and null topology
Graph exports; optional genome-linked network overlays

ASV-VOC analysis

Correlation analysis

Spearman / p-values / BH-FDR

Match ASVs, processed VOCs and sample metadata
Sample-type subsets and sample-level association tables

Clustering and mapping

Hierarchical clustering / heatmaps

VOC-ASV and VOC-profile clustermaps
Indicator overlays and focused ISA-VOC comparisons

ISA overlays

Sample-VOC analysis

Z-score summaries

Sample-level VOC patterns and annotated heatmaps
Case summaries display mean and standard deviation

Participant-VOC analysis

Mann-Whitney / permutations / BH-FDR

Aggregate within participants; RA / CLR concordance
Patient-level inference and cancer / control comparisons

Genome references
(optional)

Genome linkage

BLAST+ / genome-derived 16S references

Identity / query-coverage and taxonomy gates
Unique / ambiguous matches; genome QC metadata

Genome-linkage tables
Network overlays / module anchors

Summary reporting

(figures and tables)